

Black Holes in the Rope Framework

A Comb, Not a Point — the tension-exhaustion horizon, the percolation collapse, mass without knots, the reconnection count, and a registered divergence from Hawking. Version 2.2.7. Claims GRV-034 through GRV-039; every quantitative statement below is backed by a benchmark in benchmarks/gravity/ and registered in claims.yaml with status, dependencies, and honest limits.

1. The question, and a contradiction the corpus was carrying

For most of this framework's life, the black hole was a placeholder. The plain-language guide said the ropes near a black hole 'run out of tension almost entirely'; the cosmology and thermodynamics papers hedged with the opposite adjective, 'extreme-tension configurations'; and this paper existed as a title with no content. Nobody had derived anything, so nobody had noticed that the corpus disagreed with itself. The gravity campaign of v2.2.6–v2.2.7 — the induced Einstein–Hilbert dynamics (GRV-025), the one-metric dictionary (GRV-029), and the strand-level crossing measurements (GRV-027) — supplied, for the first time, instruments that could adjudicate.

2. The horizon is the tension-exhaustion surface (GRV-034)

The one-metric dictionary reads any static metric as a medium: $T = AB$, $\mu = B^3/A$, $c_{\text{local}} = A/B$ (with $A = \sqrt{-g_{00}}$, $B = \sqrt{g_{\text{spatial}}}$, isotropic coordinates). Evaluated along Schwarzschild, the verdict is exact and benchmarked: the effective transverse tension T falls monotonically and reaches ZERO precisely at the horizon; the effective inertia μ diverges; the local transverse wave speed c_{local} goes to zero. The horizon is not a one-way membrane in spacetime; it is the surface where the medium can no longer carry outward transverse waves — the frozen-star picture with a constitutive mechanism. The guide's amateur adjective was right; the technical hedge was wrong; both papers were corrected in place. Honest limit, stated at full volume here and in every claim of this arc: the Einstein–Hilbert dynamics were derived at weak field, so everything below is the derived dictionary EXTRAPOLATED along the classical solution, valid exactly as far as EH holds. (Audited at GRV-048: this extrapolation is a CONTROLLED expansion — the parameter is curvature-invariant times lattice scale, never the potential, and it evaluates to $1e-78$ at a stellar horizon; Ricci-class induced corrections vanish exactly on the Ricci-flat solution; the dictionary holds pointwise at ten-fold conditioning. Strong potential is not strong curvature: the drama at the horizon is about time, not about curvature.)

WHAT THIS MEANS, IN PLAIN TERMS: it does not mean the ropes go slack, and it does not mean any rope breaks — the strands of this framework are inextensible and unbreakable, and every conservation law depends on that. The individual strands near the horizon are under ENORMOUS radial tension, hauled taut toward the mass exactly as intuition demands, and that radial tension-energy is in fact the black hole's mass (Section 4). What reaches zero is the medium's capacity to deliver a transverse wave OUTWARD, and it fails through two mechanisms, neither of which severs anything. First, the FREEZE: a transverse wave can always run along a single taut strand — no crossings required — but near the horizon every oscillation is slowed by gravitational time dilation

toward a standstill as reckoned from outside; an outbound pulse on a radial strand does not hit a wall, it redshifts toward zero frequency and takes forever to climb out. Second, the DECOUPLING: the weave's lateral transmission runs through strand crossings, and the extreme radial load presses each crossing past its finite barrier so that the over-strand slips to the under side — a configuration change, not a break; the rope is intact before and after, but the over/under arrangement that handed lateral force from strand to strand is gone. Frozen along the strands, decoupled between them: that is 'exhaustion' — the tension budget consumed holding the radial load, with no outward signal transport remaining in either channel.

3. The tension paradox, and the percolation horizon (GRV-035)

Intuition objects: tension near an enormous mass should be HIGH — matter pulls. The resolution required refusing the dichotomy, and it is the arc's central mechanical insight. The radial, per-strand tension IS high: the strands are hauled taut toward the mass. But the dictionary's vanishing T is the TRANSVERSE effective tension — the wave-carrying coupling of the weave, which lives in the strand crossings — and light is a transverse wave. The corpus had already measured what extreme pressing does to a crossing: above a finite threshold the contact barrier gives way and the crossing punches through (GRV-027): the over-strand slips past to the under side — a configuration change, not a break; no rope is ever severed — deleting a transverse coupling. As conditioning intensifies, crossings fail one by one and the transverse network thins toward disconnection.

Connectivity collapse has a name: percolation. The two-stage model — crossing survival $p(T_{\text{press}})$ measured from the exact contact solver with weave disorder, composed with the transverse stiffness $K_{\text{eff}}(p)$ of the diluted network — produced an unplanned validation: K_{eff} vanishes at $p = 0.24 - 0.25$, reproducing the known three-dimensional bond-percolation threshold (0.2488) that the model was never told. The horizon is sharp because percolation transitions are sharp. Beyond it, the interior is a radially-combed forest of taut strands with no weave between them: transversely silent in exactly the channel light lives in, radially strung, continuously reconnecting. A comb, not a point. The singularity of the smooth description is a topology-change region with a mechanism — reconnection — not a mystery.

4. Mass without knots, and no-hair as strand mechanics (GRV-036)

If the interior disassembles knots, and the glossary defines mass as a knot's resistance to acceleration, how does a black hole have mass? The audit separated two definitions that coincide for particles and part company at the horizon. Knot-resistance is the PARTICLE-sector inertial definition. The gravitating source is stress-energy (GRV-026's covariant sourcing), and strand energy gravitates knotted or not: tension times length is energy, and $E = mc^2$ is the bridge. When a knot is disassembled its energy transfers — the conservative engine forbids otherwise — into the comb. The black hole's mass IS the tension energy of the comb: mass with no rest-mass carriers, the way a box of light gravitates. Verified: the far-field mass read from the (T, μ) profiles alone recovers the input to one percent, and punch-through RELEASES energy (measured $\Delta E > 0$) rather than losing it. Both halves of the original intuition land: high radial tension is the MASS; exhausted transverse tension is the HORIZON.

Pushing the audit through the survives-versus-dies ledger produced the arc's unexpected gift: the no-hair theorem's exact inventory. (For readers meeting the term for the first time: general relativity's no-hair theorem states that once a black hole settles down, it is completely characterized by just THREE numbers — mass, electric charge, and angular momentum. Everything else about what fell in — whether it was hydrogen or iron, matter or antimatter, a star or a library — is unrecoverable from outside; the hole has no other distinguishing features, no 'hair,' in John Wheeler's famous phrase. It is one of the most striking and best-established results about black holes, and any microscopic theory of what a black hole IS must explain why exactly those three properties survive and nothing else does.) What survives disassembly is precisely the strand-LOCAL conserved quantities — energy (tension-length), electric charge (the helix handedness of strand material, which reconnection rejoins but never erases), angular momentum (circulation). What dies is precisely the knot-TOPOLOGICAL quantities — baryon number, lepton number, particle identity — erased into reconnection microstates. That is general relativity's 'mass, charge, spin, and nothing else', here as the knot-versus-strand distinction; baryon non-conservation in black holes, standard lore with no mechanism in GR, has one: the baryon was a knot, and the knot is gone.

5. The reconnection count: entropy as one-way bits (GRV-037, GRV-038)

The entropy question and the information question now aim at one countable object. The microphysics: near its threshold each crossing is a two-state system, and the engine measurement delivered the crucial asymmetry — the intact branch is metastable up to $T_c \sim 1.75$ while the punched-through branch is ABSORBING at every tension tested. Reconnection is a one-way ratchet: the count is monotone, which is Hawking's area theorem as crossing mechanics, and the erased identity of Section 4 is frozen into the through-pattern — not destroyed, unread.

The bits live on the reconnection-active shell, where the disorder-broadened transition band sits. Whether that shell's population scales with area or volume hung on one bridge: the profile of pressing versus distance. GRV-038 delivered it with one premise, one exact theorem, and one dispatch. The premise is the corpus's own picture stated as input: the frozen medium outside the horizon is static and bears its static support load, proportional to local proper acceleration. The theorem is exact: $a(r) s(r) \rightarrow c^2$ as the horizon is approached, at every mass — verified as a thirteen-decimal collapse of the curves for $r_s = 1, 10, 100$ onto one universal function, with mass-dependence entering only at $O(s/r_s)$, which at the shell is of order $1e-50$ for astrophysical holes. The dispatch: the tidal alternative — the only profile class that would have killed the area law — is suppressed by $(a_{\text{cell}}/r_s)^2$. The shell therefore has fixed thickness; the count scales with AREA (lattice demonstration: exponent 1.9 across a tripling of horizon radius); and $S \sim A/a^2$ meets GRV-007's registration, from two eras earlier and for independent reasons, of the gravitational cutoff at the Planck length: the Bekenstein-Hawking FORM, $S \sim A/l_P^2$, assembled from named parts. The order-one coefficient is not claimed — for the reason Section 6 makes precise.

6. The Hawking channel: birth-cry, then silence (GRV-039)

The temperature question was put to a formation experiment: exact Gaussian (covariance-matrix) quantum dynamics of the transverse chain, initial uniform-medium vacuum, tension profile ramped

to the exhaustion configuration with a linear $c(x)$ crossing of surface gravity $\kappa = dc/dx$, then free evolution with detectors downstream. Two results, both measured. First: the collapse RADIATES — a directed outgoing burst (right-mover excess over the left-mover control) carrying the formation's quantum imprint. Second: once the burst passes, the steady current settles to about one percent of the one-dimensional Stefan-Boltzmann prediction at $T = \kappa/2\pi$, and stays there.

The mandatory audit asked whether the null is the lattice's fault, and concluded it is the physics: steady Hawking flux is powered by continuous vacuum conversion at an infinite-redshift surface, or by a supply flow in analogue systems — and the corpus's horizon is NEITHER. It is a percolation-severed transverse channel with a strand-scale floor: Boulware-like at late times. The medium does not merely fail to reproduce the standard mechanism; its derived structure removes it. House rules leave one path: register the divergence and arm its falsifier. This branch predicts that black holes emit when MADE and when FED — formation bursts, and accretion-driven reconnection noise from the marginal shell, each ratchet event releasing its measured quantum of energy — and that isolated holes are eternally quiet. For every observed astrophysical hole this is indistinguishable from Hawking (whose $1e-8$ K is unobservable regardless); for primordial black holes it is a sharp fork: A DETECTED PRIMORDIAL-BLACK-HOLE EVAPORATION BURST WOULD FALSIFY THIS BRANCH. The famous $1/4$ was checked as arithmetic — $T = \kappa/2\pi$ with the first law integrates to $S = A/4$ on the nose — and declined as a claim, because the corpus rejects the premise at the steady level. A framework that manufactures the most celebrated coefficient in the field would be worth less than one that declines it for stated reasons.

7. The whisper: Hawking's radiation, rebuilt from rope (GRV-040 ... GRV-044)

Section 6 ended on a divergence and a named summit: characterize the reconnection noise. One cycle later the summit is a five-claim arc, and it delivered more than the conjecture asked. THE SCALE (GRV-040): a ratchet event at the shell emits at strand-scale frequency from strand-scale depth, and the Rindler redshift factor $\kappa \cdot s$ makes the strand scale CANCEL -- any microscopic emission from its own natural depth arrives at infinity at the horizon's own frequency. The $O(1)$ coefficient, built entirely from engine quantities measured for other purposes (the punched-through crossing's ringdown, 0.35; the survival-weighted shell depth, 0.65), came out at 0.23κ -- within a factor of two of Hawking's own spectral peak at 0.45κ . The escape was verified by exact transfer matrix (transmission 4-11 percent through the exhaustion gradient), and the luminosity law closed under energetics: $L = f (dM/dt) c^2 T_{\text{grey}}$ with f bounded by one -- accretion-powered, so the isolated silence of Section 6 is preserved exactly.

THE SPECTRUM (GRV-041/042): does the ratchet ensemble thermalize? Three lattice generations answered with a convergence and a disciplined refusal: the ensemble preferred the Planck form over a power law at every resolution, the two independent temperature estimators agreed, and the locked exponential-tail bar (r -squared above 0.98) failed at every finite lattice -- with the failure diagnosed: discrete depth grids truncate the deep-redshift pile-up whose completion is the exponential. The registry therefore recorded a trend, a mechanism, and an apparent limit, and refused the word 'thermal.'

THE CONTINUUM (GRV-043): the analytic pile-up integral -- every ingredient independently measured (the intrinsic burst kernel from a flat-chain quench, Gaussian-tailed; the depth density from the survival curve; the greybody from the transfer function) -- passed the bar the lattices could not (r -squared 0.983) and delivered two results of theorem flavor. First, GEOMETRY THERMALIZES: the depth integral converts the Gaussian kernel into a thermal-class envelope -- the redshift pile-up does not merely hide the microphysics, it transmutes it. Second, THE TEMPERATURE: the Planck fit landed at 0.21 kappa against Hawking's kappa-over-two-pi of 0.16 -- a ratio of 1.3, from a kernel, a survival curve, and a transmission function none of which knows what two-pi is. The offset is recorded as unexplained; no exact claim is made.

THE TAIL'S TRUE ANATOMY (GRV-044, Derived): boundary-Laplace analysis of the integral gives the closed form $F(\omega)$ proportional to $\omega^{-2} \exp(-\lambda \omega^2 / (\kappa s^2)^2)$, verified against the exact integral to five percent -- and it AMENDS the numerical classification: the true asymptotic tail is GAUSSIAN, super-exponential, with a RUNNING effective temperature T_{eff} proportional to $1/\omega$; the 'exponential' of the continuum fit was the finite-window effective slope, and the window-averaged temperature computed from the closed form (0.209 kappa) reproduces the fitted 0.21 exactly. The pattern that produced this arc deserves its sentence: the numerics propose, the bar restrains, the derivation disposes.

THE OBSERVATIONAL LEDGER, three independent discriminators between strand black holes and textbook ones: (1) isolated holes are silent -- a detected primordial-black-hole evaporation burst falsifies this branch (Section 6); (2) emission is GATED by feeding -- a switch, not a slope (GRV-047): the whisper saturates at a channel-limited ceiling of $5.8e-4$ times the Hawking power, mass-independently, at roughly 180 Hz for a ten-solar-mass hole and in the LISA band for Sagittarius A* -- Hawking-faint, with no detection claim made; (3) the tail runs cold -- Hawking's temperature is constant while the whisper's falls as $1/\omega$, leaving the spectrum tail-deficient relative to a matched Planck curve. Measure the tail, and the power source identifies itself. A framework that declined Hawking's mechanism has rebuilt his radiation's every public feature -- scale, family, temperature to thirty percent -- and knows exactly where the pictures part.

8. Honest limits and the open program

Everything above inherits the arc's shared caveat: the weak-field-derived dictionary extrapolated to strong field. Additional named limits: the load-share constant K (the familiar absolute-scale problem -- it never touches a scaling result); the mesoscale character of the percolation and counting models (scalar-displacement network, single crossing geometry, stylized formation profile); the bit-identification via the mixed-population rule; and the entropy coefficient, open. The named next-orders, in priority order: the temperature-offset anatomy (does the 1.3-times-kappa-over-two-pi offset resolve analytically from kernel width and greybody detail?); the f -efficiency computation from accretion-driven crossing turnover; the Weyl-squared coefficient from the weave band (the strong-field audit is DONE -- GRV-048 opened the gate -- and this coefficient would convert its one named premise to a measurement); the three-dimensional greybody; and the handedness-through-reconnection benchmark that would promote the no-hair correspondence to Derived.

Registered claims: GRV-034 (tension-exhaustion horizon), GRV-035 (percolation collapse), GRV-036 (mass without knots; no-hair), GRV-037 (the reconnection count), GRV-038 (the pressing profile), GRV-039 (the Hawking channel; the registered divergence), GRV-040 (the whisper's scale, escape, and luminosity law), GRV-041 (the shape verdict), GRV-042 (the tail convergence), GRV-043 (the pile-up integral; the temperature), GRV-044 (the saddle-point derivation; the running tail -- Derived). Dependencies reach GRV-005, GRV-007, GRV-025-029, and FND-KIN-005. A black hole, in this corpus: a comb that remembers its area, announces its birth, whispers when it eats, and otherwise keeps perfect silence.